

■ SQL Revision Notes: The Simple SELECT Statement

1. What is the SELECT Statement?

The SELECT statement is the most fundamental command in SQL. Its purpose is simple: to retrieve (query) data from a database and return it as a result set (table).

Concept	Explanation	Simple Analogy
SELECT	Specifies which columns (or data points) you want to see.	Like choosing which shelf (column) to look at in a library.
FROM	Specifies where the data is located (the table name).	Like choosing which room (table) in the library to enter.
Result Set	The new table of data returned after running the query.	The pile of books (rows/columns) you take out of the library.

2. Retrieving Data (Basic Syntax)

A. Retrieving ALL Columns

Command	Example	Goal
SELECT * FROM TableName;	SELECT * FROM Employees;	Returns ALL columns and ALL rows from the "Employees" table.

B. Retrieving SELECTED Columns

	Example	Goal
Column2 FROM TableName;	SELECT FirstName, LastName, Email FROM Customers;	Returns only the FirstName, LastName, and Email from the Customers table.

C. Retrieving Unique Values (DISTINCT)

	Example	Goal
DISTINCT Column FROM TableName;	SELECT DISTINCT City FROM Stores;	Shows a list of every unique city where a store is located.

3. Filtering Results (WHERE Clause)

The WHERE clause is a criteria command used to filter the rows returned by your query. It's the most common way to narrow down your results.

A. Basic Syntax

```
SELECT Column1, Column2
FROM TableName
WHERE Column = 'Criteria';
```

B. Common Filtering Operators

Operator	Meaning	Example
=	Exactly equal to (case-sensitive for text)	WHERE Country = 'Canada'
!= or <>	Not equal to	WHERE JobTitle <> 'Manager'
>	Greater than (for numbers/dates)	WHERE Salary > 50000
<	Less than (for numbers/dates)	WHERE HireDate < '2023-01-01'

>= / <=	Greater/Less than or equal to	WHERE Age >= 18
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C. Combining Filters (Logical Operators)

AND: BOTH conditions must be true.

Example: WHERE Country = 'USA' AND City = 'New York'

OR: AT LEAST ONE condition must be true.

Example: WHERE JobTitle = 'Analyst' OR JobTitle = 'Trainee'

4. Key Takeaways & Practice Tips

- **Order Matters:** In a simple query, the keywords SELECT and FROM must appear in that order.
- **Semicolon:** Always end your statement with a semicolon (;).
- **Use Quotes:** Text values (strings) in the WHERE clause usually require single quotes (e.g., 'USA'). Numbers do not (e.g., 50000).
- **Readability:** Write your keywords (SELECT, FROM, WHERE) in capital letters to make your SQL code easier to read.
- **Troubleshooting:** If your query returns a syntax error, check your column and table names for typos first, then verify your operator usage.

Practice Tip: Try writing one query for each concept without looking at the solutions. That's the best way to make these concepts stick! ■